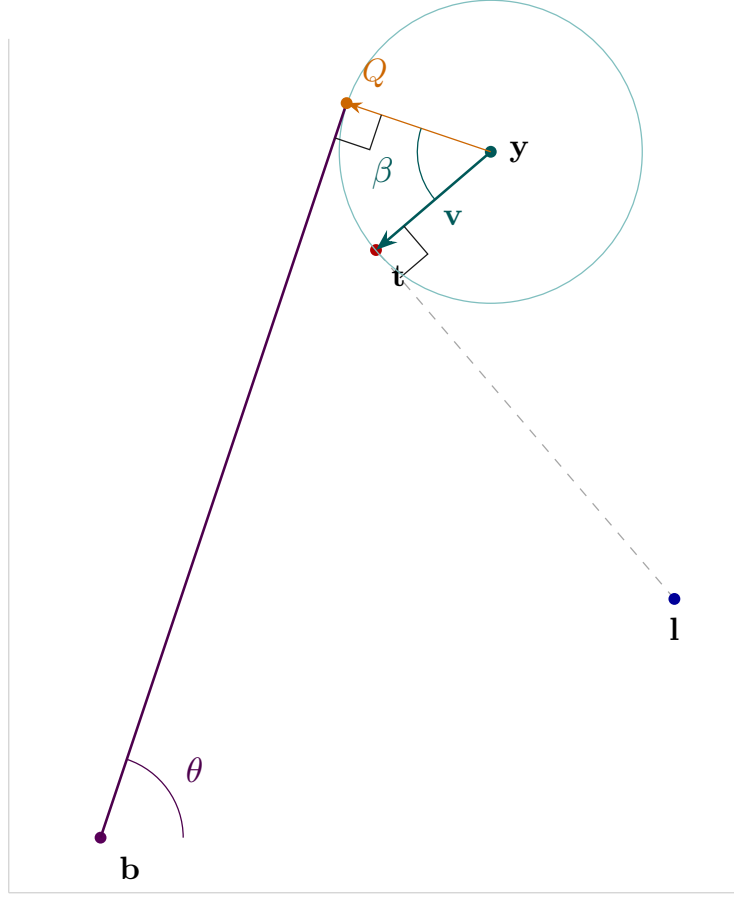


IKCompiler.calc_yaw_and_base_angles()



$\mathbf{v} \perp \mathbf{lt}$ at $\mathbf{t}$. $\mathbf{bQ} \perp \mathbf{yQ}$ at Q – the tangency condition, Eq. (4). Teal circle: radius L_2 about $\mathbf{y}$.

1 Setup

$\mathbf{b} = \text{base_position}, \quad \mathbf{l} = \text{lens.position}, \quad \mathbf{t} = \text{target.pos} \quad \in \mathbb{R}^3$
 $L_2 = \text{L2}$

Only the xy -projections matter below; write $\mathbf{b}, \mathbf{l}, \mathbf{t}, \mathbf{y}$ for them from here on.

2 Yaw Axis Position

$$\mathbf{d} = \mathbf{t} - \mathbf{l}, \quad \hat{\mathbf{n}} = \frac{(-d_y, d_x)}{\|\mathbf{d}\|} \quad (1)$$

$$\mathbf{v} = L_2 \hat{\mathbf{n}}, \quad \mathbf{y} = \mathbf{t} - \mathbf{v} \quad (2)$$

3 Tangent Condition

$$\mathbf{v}_r(\beta) = R(\beta) \mathbf{v}, \quad Q(\beta) = \mathbf{y} + \mathbf{v}_r(\beta), \quad R(\beta) = \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix} \quad (3)$$

$$\mathbf{v}_r(\beta) \cdot (Q(\beta) - \mathbf{b}) = 0 \quad (4)$$

radius $\mathbf{y}Q(\beta) \perp$ tangent $\mathbf{b}Q(\beta)$: $\mathbf{v}_r(\beta)$, the rotated offset, is perpendicular to the line from the base to the point it lands on.

4 Solving (4)

Write $\mathbf{w} = \mathbf{y} - \mathbf{b}$. Since $\|\mathbf{v}_r(\beta)\| = \|\mathbf{v}\| = L_2$ for every β :

$$\mathbf{v}_r(\beta) \cdot (Q(\beta) - \mathbf{b}) = \mathbf{v}_r(\beta) \cdot (\mathbf{w} + \mathbf{v}_r(\beta)) = \mathbf{v}_r(\beta) \cdot \mathbf{w} + L_2^2 = 0 \quad (5)$$

$$\implies \mathbf{v}_r(\beta) \cdot \mathbf{w} = -L_2^2 \quad (6)$$

Substitute $\mathbf{v}_r(\beta) = (v_x \cos \beta - v_y \sin \beta, v_x \sin \beta + v_y \cos \beta)$ and group by $\cos \beta, \sin \beta$:

$$\mathbf{v}_r(\beta) \cdot \mathbf{w} = (v_x \cos \beta - v_y \sin \beta) w_x + (v_x \sin \beta + v_y \cos \beta) w_y \quad (7)$$

$$= \underbrace{(v_x w_x + v_y w_y)}_A \cos \beta + \underbrace{(v_x w_y - v_y w_x)}_B \sin \beta \quad (8)$$

i.e. $A \cos \beta + B \sin \beta = C$ with

$$A = v_x w_x + v_y w_y, \quad B = v_x w_y - v_y w_x, \quad C = -L_2^2. \quad (9)$$

5 Closed Form

Divide $A \cos \beta + B \sin \beta = C$ through by $R := \sqrt{A^2 + B^2}$:

$$\frac{A}{R} \cos \beta + \frac{B}{R} \sin \beta = \frac{C}{R} \quad (10)$$

$(A/R, B/R)$ is a unit vector, so it is $(\cos \varphi, \sin \varphi)$ for some φ – read off directly as $\varphi = \text{atan2}(B, A)$:

$$\cos \varphi = \frac{A}{R}, \quad \sin \varphi = \frac{B}{R} \quad (11)$$

Substituting turns the left side into the cosine angle-difference identity:

$$\cos \varphi \cos \beta + \sin \varphi \sin \beta = \cos(\beta - \varphi) = \frac{C}{R} \quad (12)$$

Real solutions exist iff $|C| \leq R$ (else $\mathbf{b}$ is unreachable by a tangent from this circle). Isolating β :

$$\beta - \varphi = \pm \arccos\left(\frac{C}{R}\right) \quad (13)$$

$$\boxed{\beta = \varphi \pm \arccos\left(\frac{C}{R}\right)} \quad (14)$$

6 Base Angle

$$Q = \mathbf{y} + \mathbf{v}_r(\beta), \quad \theta = \text{atan2}(Q_y - b_y, Q_x - b_x) \quad (15)$$

The two signs give two (β, θ) pairs; duplicates are dropped and the pair closest to world heading 0° is returned first.

7 Notation $\leftrightarrow$ Code

Symbol	Variable
$\mathbf{b}$	<code>base_position</code> / <code>base_xy</code>
$\hat{\mathbf{n}}$	<code>lens_2.target_perpendicular_vec</code>
$\mathbf{v}$	<code>l2.translation_vec</code>
$\mathbf{y}$	<code>yaw_axis_position</code>
A, B, C	<code>A</code> , <code>B</code> , <code>C</code>
R, φ	<code>R</code> , <code>phi</code>
β	<code>beta</code>
$\mathbf{v}_r$	<code>v_r</code>
Q	<code>q</code>
θ	<code>base_angle</code>
(β, θ) pairs	<code>solutions</code> / <code>unique_solutions</code>